ROSE CRISP

Senior Engineer & Technical Educator | AI Infrastructure on Kubernetes

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US/Taiwan dual citizen | English, Mandarin, Spanish (fluent) | Open to relocation within the US and to travel

SUMMARY

Engineer-educator with 25+ years in software and infrastructure automation, including 6 years at Red Hat building, validating, and teaching cloud-native and AI platforms. Stands up GPU-backed AI stacks on Kubernetes/OpenShift, from vLLM inference on OpenShift AI to RAG and agent apps, and automates them with Go operators, Helm, and GitOps. Turns working deployments into hands-on workshops, demos, and conference talks.

TOP SKILLS

- Kubernetes & Red Hat OpenShift
- Kubernetes Operators in Go (Operator SDK, OLM)
- LLM inference serving: vLLM & KServe
- Red Hat OpenShift AI (GPU & CPU)
- RAG & AI agents (Llama Stack)
- Python
- Helm & GitOps (Argo CD)
- Terraform & Ansible automation
- Observability: Prometheus metrics & alerting
- Technical workshops, demos & conference talks

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Red Hat, Raleigh, NC | 02/2020 - Present

Senior Software Engineer and Educator; partner engineering for OpenShift and OpenShift AI.

- Design and deploy proof-of-concept AI workloads on OpenShift, including GPU-backed RAG chatbots and LLM serving with KServe and vLLM on OpenShift AI.
- Partner with ISVs to get their applications certified and listed in the Red Hat Ecosystem Catalog, from the first proof of concept through certification.
- Contribute to and run the Red Hat Best Practices for Kubernetes test suite (certsuite). I add new test cases and validate 50+ partner operators on every new OpenShift release, alerting stakeholders when an operator doesn't pass certification.
- Lead Red Hat's Kubernetes Operator Framework workshops, training partners to build, package, and certify operators.
- Built a Red Hat Developer Hub (Backstage) golden-path template that automates GitOps repo scaffolding, Argo CD, and Helm deployments for GPU-backed RAG workloads, turning a multi-team manual setup into a self-service flow (DevConf.US 2025).

Senior Software Developer | SAS Institute Inc., Cary, NC | 04/2019 - 12/2019

- Built infrastructure automation with Ansible, Terraform, Bash, HashiCorp Vault, and OpenStack, and CI/CD pipelines in Jenkins and GitLab with automated tests and security scans.
- Containerized application deployments on Kubernetes and Docker, managed with Helm charts.

Senior Software Developer | SAS Institute Inc., Cary, NC | 05/2015 - 04/2019

- Automated the SAS software escrow process with Jenkins, Perl, Bash, and MySQL, increasing production by 400%.
- Managed 600+ N+1 software feature releases per year, from infrastructure setup to delivery.

Earlier roles | SAS Institute Inc., Cary, NC | 06/1994 - 05/2015

- Software Developer, Source Management; Software Tester, Source Management; Associate Systems Administrator (1996-1997); Technical Student (1994-1996).

TALKS & WORKSHOPS

- DevConf.US 2025 - "Zero to GitOps: Automating RAG Workloads with Backstage Templates"
- DevConf.CZ 2024 - Workshop co-lead, "Developing your Level 5 Operator in 75 Minutes"
- Cloud Native Rejekts EU 2022 - "From Zero to Auto Pilot: Exploring Kubernetes Operator Capability Levels"
- THAT Conference, PlatformCon, DeveloperWeek Europe, WomenTech Global 2023 - "Create Your Own Resources with CRDs"; Config Management Camp 2024 - Ansible on AWS

SELECTED PROJECTS (GITHUB.COM/ROCRISP)

- llm-cpu-serving (Red Hat AI quickstart, contributor) - vLLM inference via KServe on OpenShift AI, with Helm charts and architecture docs.
- aps-rag - RAG reference architecture with Llama Stack, OpenShift AI, PGVector, and Kubeflow Pipelines.
- Go Kubernetes operators - openedx-operator, iot-operator, admission webhook and more, built with Operator SDK and packaged for OLM.

EDUCATION

North Carolina State University, Raleigh, NC - Computer Science coursework toward a B.S. (127 credits completed), 1989-1992